

Zihan Ding

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EDUCATION

- Ph.D. in Computer Science, Stony Brook University** Aug 2023 – Present
– Research: AI for healthcare, LLMs for clinical notes, multimodal EHR model design. Advisor: [Fusheng Wang](#)
- M.S. in Computer Science, Stony Brook University** Aug 2021 – May 2023
- B.E. in Computer Science and Technology, Anhui University** Sep 2016 – Jun 2020

PUBLICATIONS

MANUSCRIPTS UNDER REVIEW

- Zihan Ding**, Grace Han, Tengfei Ma, Fusheng Wang, Janos Hajagos, Tahsin Kurc, and Anisha R. Kumar (2027). “From National Data to Local Evaluation: Benchmarking Machine Learning Models for Perioperative Risk Prediction”. Submitted to the 2027 AMIA Amplify Informatics Summit.
- Xiyue Jiang, **Zihan Ding**, Grace Han, Yinan Liu, Richard N. Rosenthal, and Fusheng Wang (2027). “Beyond the EHR: Patient-Reported Survey Data Improve Prediction of Opioid Use Disorder”. Submitted to the 2027 AMIA Amplify Informatics Summit; co-first author, equal contribution.
- Yinan Liu, **Zihan Ding**, Tengfei Ma, Rachel Wong, Xia Zhao, Xia Zheng, George S. Leibowitz, Ryan DiMeglio, Cari Faith Besserman, Kelly S. Ramsey, Charlene Chen, Richard N. Rosenthal, and Fusheng Wang (2027). “Measuring Stakeholder-Driven Model Change: A Configurable Pipeline to Support Stakeholder-in-the-Loop Clinical AI”. Submitted to the 2027 AMIA Amplify Informatics Summit.

JOURNAL ARTICLES

- Zihan Ding**, Jianyuan Deng, Linda Mermelstein, Barbara Nemesure, Tyler Osborne, and Fusheng Wang (2025). “Demographic disparities, temporal trends, and geographic patterns of HPV vaccination on Long Island, New York: A comprehensive analysis of immunization registry data (2012–2023)”. In: *Human Vaccines & Immunotherapeutics* 21.1, p. 2487383. DOI: [10.1080/21645515.2025.2487383](https://doi.org/10.1080/21645515.2025.2487383).
- Yuanting Yan, Ruiqing Liu, **Zihan Ding**, Xiuquan Du, Jie Chen, and Yanping Zhang (2019). “A Parameter-Free Cleaning Method for SMOTE in Imbalanced Classification”. In: *IEEE Access* 7, pp. 23537–23548. DOI: [10.1109/ACCESS.2019.2899467](https://doi.org/10.1109/ACCESS.2019.2899467).

PEER-REVIEWED CONFERENCE PAPERS

- Zihan Ding**, Yinan Liu, Tengfei Ma, Rachel Wong, Xia Zhao, Richard N. Rosenthal, and Fusheng Wang (2026). “A Comparative Study of Feature Selection Methods for EHR Diagnosis Codes in Opioid Use Disorder Prediction”. In: *2026 AMIA Annual Symposium Proceedings*. To appear. arXiv: [2608.04180](https://arxiv.org/abs/2608.04180) [cs.LG].
- Zihan Ding**, Xinyu Dong, Yinan Liu, Tengfei Ma, Xia Zhao, Rachel Wong, Richard N Rosenthal, and Fusheng Wang (2025). “HIBERT: A Hybrid Clustering BERT for Interpretable Opioid Overdose Risk Prediction”. In: *AMIA Annual Symposium Proceedings*. Vol. 2025, p. 303.
- Yukun Yuan, **Zihan Ding**, and Shan Lin (2024). “Fairness-Aware Electric Taxi Fleet Coordination Under Short-Term Power System Failures”. In: *2024 American Control Conference (ACC)*. IEEE, pp. 214–219. DOI: [10.23919/ACC60939.2024.10645001](https://doi.org/10.23919/ACC60939.2024.10645001).
- Zihan Ding**, Ruting Rao, Yuanting Yan, and Yanping Zhang (2019). “Voting Based Constructive Covering Algorithm”. In: *2019 IEEE 14th International Conference on Intelligent Systems and Knowledge Engineering (ISKE)*. IEEE, pp. 720–724. DOI: [10.1109/ISKE47853.2019.9170310](https://doi.org/10.1109/ISKE47853.2019.9170310).

GRANTS AND FUNDING

- Climate Change Seed Grant, Stony Brook University** — funded 2024
- Project: *Building a Comprehensive Repository of Health and Climate Databases to Study the Impact of Climate Change on Mental Health with a Pilot Study on Opioid Overdose*.
- Contributed to the successfully funded proposal through literature review, preliminary analysis, and scientific writing.

RESEARCH EXPERIENCE

HUMAN-CENTERED AI FOR CLINICAL DECISION SUPPORT

Opioid Use Disorder Risk Prediction — PCORI-funded Research Project

Aug 2024 – Present

- Work within a Patient-Centered Outcomes Research Institute (PCORI) project in which patients and clinicians are embedded throughout the research process, co-defining prediction targets, clinical features, and evaluation criteria rather than receiving a finished model.
- Designed and built the clinical decision support dashboard that lets clinicians compare patient risk across trained models and customize cohorts, features, diseases, and other clinical variables, iterating directly with clinical stakeholders.
- Audited every open and proprietary dataset available to the project for its ability to support Morphine Milligram Equivalent (MME) computation, documenting how far the calculation can be carried and which fields must be derived, as a decision aid for jointly choosing a training corpus on grounds including population coverage.
- Proposed HIBERT, a two-level hierarchical transformer that encodes each clinical encounter, aggregates encounters into a patient-level representation, and clusters those representations into interpretable risk groups, with DiCE counterfactual explanations (AMIA 2025); compared clinical feature selection methods for OUD prediction (AMIA 2026); co-developed a configurable pipeline measuring how a model changes when stakeholders revise its cohort, features, or targets (under review).

Multimodal Opioid related Disease Early Risk Prediction

Aug 2025 – Present

- Building a **multimodal transformer architecture** for opioid-related risk prediction that fuses structured EHR data with unstructured clinical notes at the hidden-state level, incorporating decoder-only LLMs so the model returns not only a risk score but the clinical rationale behind it, making predictions auditable by clinicians rather than opaque.
- Process notes through multiple **locally deployed LLMs** to strip sensitive information and restructure free text into tabular form, keeping protected health information on-premise while exploiting the language understanding of LLMs to assist multimodal fusion.
- Currently designing a **clinical note selection framework** that identifies which notes in a patient's longitudinal record carry decision-relevant signal, alongside EHR alignment methods that temporally align notes with structured records, reducing the volume clinicians and models must process while making each prediction traceable to specific documentation.

Multimodal Perioperative Risk Prediction using Surgical Notes and EHR Data

Jan 2026 – Present

- Built multimodal learning pipelines integrating structured EHR data with longitudinal surgical notes, using transformer-based note fusion to capture temporal relationships among preoperative, intraoperative, and postoperative documentation, and evaluated LLM-based representations and structured-to-text serialization for predicting postoperative mortality, readmission, and reoperation.
- Benchmarked models trained on a national surgical registry against evaluation at a single institution, characterizing how far nationally trained perioperative risk models transfer to local practice (first author, under review).

POPULATION HEALTH AND HEALTH DISPARITIES

GeoSpatial analysis of Long Island HPV vaccine uptake

Aug 2023 – Oct 2024

- Analyzed 2012–2023 HPV vaccination records from a statewide immunization registry (NYSIIS) and SPARCS covering 1.12M patients, building the record-linkage pipeline and ACIP/CDC-compliant dose-completion measures, then quantifying disparities and temporal trends (a 284% rise among ages 9–13 from 2012–2019, a COVID-related dip in 2020, only 25% completing the second dose on schedule) and running ZIP code-level GIS analysis that identified a persistent east–west uptake gradient and low-coverage regions for targeted outreach (first author, *Human Vaccines & Immunotherapeutics*).

Climate Change and Opioid Overdose Research Initiative

Sep 2023 – Sep 2024

- Designed a population-scale data repository integrating SPARCS health records with climate and geospatial datasets to study climate-related mental health outcomes, and produced the preliminary analyses and literature synthesis that anchored the funded seed grant proposal.

EARLIER RESEARCH: METHODS AND INTERACTIVE SYSTEMS

Fairness-Aware Coordination for Electric Taxi Fleets under Power Disruptions Aug 2022 – Aug 2023

- Proposed Fairness-Aware e-taxi fleet Coordination (FAC), a receding-horizon controller that switches between city-wide performance optimization and joint fairness–performance optimization when charging capacity goes offline; evaluated on a citywide dataset of ~8,000 vehicles (ACC 2024).

Oversampling and Ensemble Methods for Imbalanced Data Classification Jan 2018 – Nov 2019

- Proposed a parameter-free data cleaning method for SMOTE oversampling that removes hard-to-learn synthetic samples using mutual-information measures and sample space distribution, and V-CCA, a voting ensemble addressing instability under random initialization; evaluated on 25 imbalanced datasets and 16 UCI benchmarks (*IEEE Access* 2019; ISKE 2019, first author).

Interactive Visual Analytics Dashboard for Portfolio Optimization Jan 2022 – Jun 2022

- Designed and implemented a coordinated multiple-view dashboard (D3.js client, Python server) linking a geographic map, a risk–return frontier, tables, and pie charts, where a selection in any view brushes and filters the others, shifting the interaction model from a single static optimization result to user-driven what-if exploration.

MENTORING AND SUPERVISION

Undergraduate Researcher Sep 2025 – Present

- Supervise a study on multi-horizon Opioid Use Disorder risk prediction using the *All of Us* Research Program cohort, measuring how much predictive value patient-reported survey data adds over chart-derived features at 3-, 6-, 12-, and 24-month prediction windows. Survey data improved prediction at every horizon; manuscript under review with the student as co-first author.

High School Student Researcher Summer 2026

- Mentored a student in the Computer Science and Informatics Research Experience (CSIRE) summer program on an independent project introducing sequential models for opioid use disorder prediction and alignment across heterogeneous EHR data sources, guiding problem formulation, model selection, and data preparation.

AWARDS AND HONORS

Individual Scholarship, Anhui University Sep 2018, Sep 2019

EXPERIENCE

Research Assistant at Stony Brook University Aug 2022 – Present

- Research assistant in Professor Fusheng Wang’s biomedical informatics group; collaborate with clinicians, epidemiologists, and public health researchers at Stony Brook Medicine and the Stony Brook Cancer Center, translating clinical questions into machine learning formulations and communicating model behavior back to non-technical stakeholders.

Graduate Teaching Assistant at Stony Brook University Jan 2022 – Jun 2022

- Courses: CSE351 Data Science Fundamentals; CSE527 Introduction to Computer Vision; CSE581 Computer Science Fundamentals: Theory.

Software Development Engineer at China Guangfa Bank Jun 2020 – Jun 2021

- Designed and developed back-end (Spring Boot) and front-end (Vue.js) for two large projects letting fund managers manage assets and evaluate portfolio risk; took a new investment product from concept through testing and launch.

Product Manager Internship at IFLYTEK Dec 2019 – Jun 2020

SKILLS

Languages:	Python, Java, JavaScript, SQL, HTML/CSS
Machine Learning:	PyTorch, Transformers, LLM fine-tuning and local deployment, scikit-learn, multimodal architectures, model interpretability (DiCE, SHAP)
Interactive Systems:	D3.js, Vue.js, Node.js, Flask, ArcGIS, clinical dashboard design
Data and Systems:	PySpark, ClickHouse, MySQL, Oracle, Git, EHR pipelines (SPARCS, NYSIIS, <i>All of Us</i>)